

Set 11

Product Category Analysis

Dataset: Product Categories

Category Sales (in \$)

Electronics 50000

Clothing 35000

Appliances 40000

1. Generate a funnel chart to analyze the sales conversion process for each product category. Label the stages and title the chart.

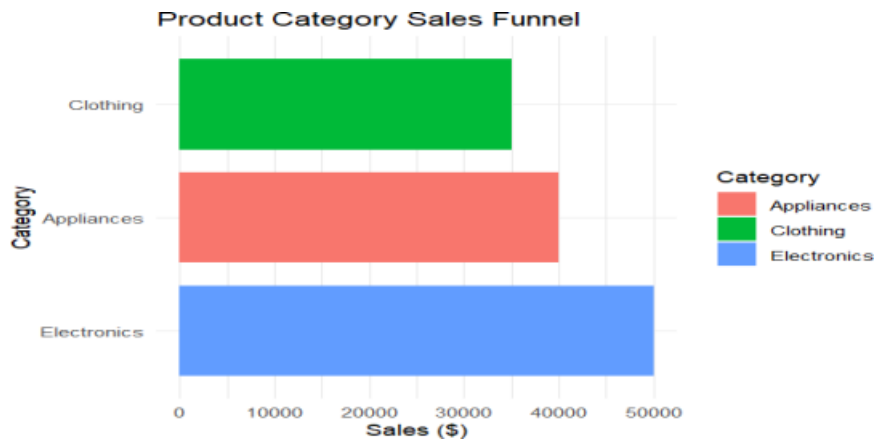
```
# Load library
library(ggplot2)

# Create dataset
category <- data.frame(
  Category = c("Electronics", "Clothing", "Appliances"),
  Sales = c(50000, 35000, 40000)
)

# Funnel Chart
ggplot(category,
  aes(x = reorder(Category, -Sales),
    y = Sales,
    fill = Category)) +
  geom_bar(stat = "identity", width = 0.8) +
  coord_flip() +
  labs(
    title = "Product Category Sales Funnel",
    x = "Category",
    y = "Sales ($)"
```

```
) +
```

```
theme_minimal()
```



2. Build a table to display the sales data for each product category. Label the table elements.

```
# Create dataset
```

```
category <- data.frame(
```

```
Category = c("Electronics", "Clothing", "Appliances"),
```

```
Sales = c(50000, 35000, 40000)
```

```
)
```

```
# Display table
```

```
print(category)
```

```
Category Sales
```

```
1 Electronics 50000
```

```
2 Clothing 35000
```

```
3 Appliances 40000
```

3. Create a pie chart to represent the distribution of sales across product categories. Include labels.

```
# Load library
```

```
library(ggplot2)
```

```
# Create dataset
```

```
category <- data.frame(
```

```
Category = c("Electronics", "Clothing", "Appliances"),
```

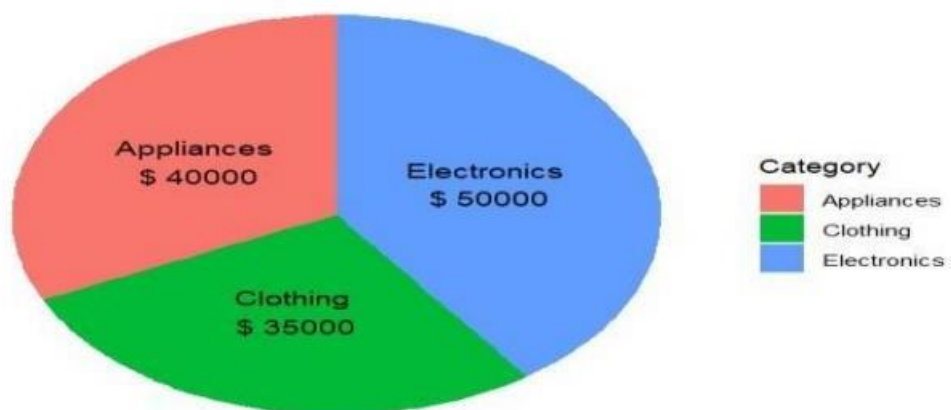
```
Sales = c(50000, 35000, 40000)
```

```
)
```

```
# Pie Chart

ggplot(category, aes(x = "", y = Sales, fill = Category)) +
  geom_bar(stat = "identity", width = 1) +
  coord_polar(theta = "y") +
  geom_text(
    aes(label = paste(Category, "\n$", Sales)),
    position = position_stack(vjust = 0.5),
    size = 4
  ) +
  labs(
    title = "Distribution of Sales Across Product Categories"
  ) +
  theme_void()
```

Distribution of Sales Across Product Categories



Set 12 :Dataset: Website Traffic

Date Page Views Click-through Rate

2023-01-01 1500 2.3%

2023-01-02 1600 2.7%

2023-01-03 1400 2.0%

1. Create a line chart to visualize the trend in daily page views over time. Label the axes and title the chart.

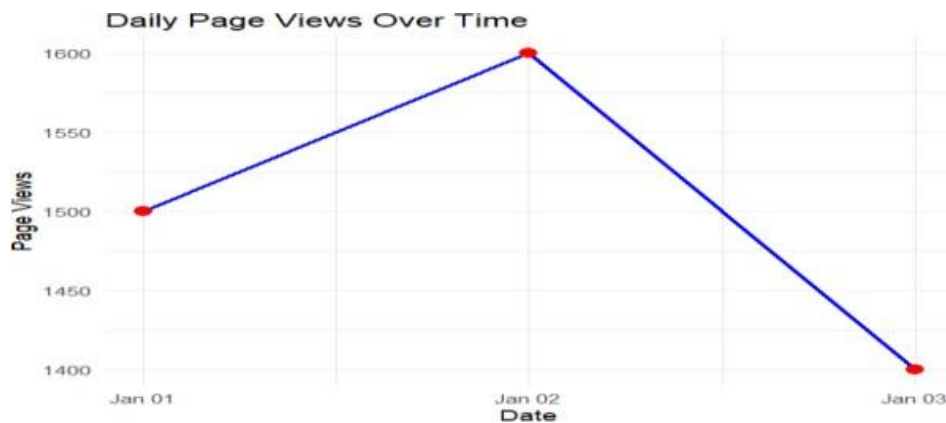
```

# Load library
library(ggplot2)

# Create dataset
traffic <- data.frame(
  Date = as.Date(c("2023-01-01",
    "2023-01-02",
    "2023-01-03")),
  Page_Views = c(1500,1600,1400),
  Click_Through_Rate = c(2.3,2.7,2.0)
)

# Line Chart
ggplot(traffic,
  aes(x = Date,
    y = Page_Views,
    group = 1)) +
  geom_line(color = "blue", linewidth = 1) +
  geom_point(color = "red", size = 3) +
  labs(
    title = "Daily Page Views Over Time",
    x = "Date",
    y = "Page Views"
  ) +
  theme_minimal()

```



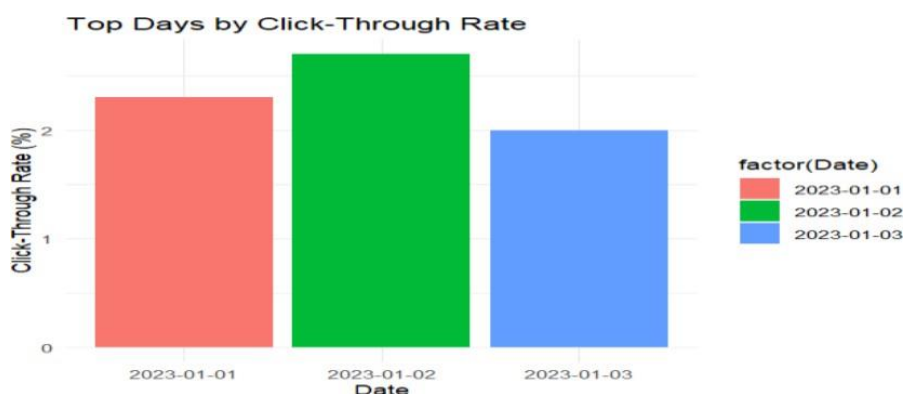
2. Generate a bar chart showing the top N days with the highest click-through rates. Label the chart elements.

```
# Load library
library(ggplot2)

# Create dataset
traffic <- data.frame(
  Date = as.Date(c("2023-01-01",
    "2023-01-02",
    "2023-01-03")),
  Click_Through_Rate = c(2.3,2.7,2.0)
)

# Sort data
top_days <- traffic[order(-traffic$Click_Through_Rate), ]

# Bar Chart
ggplot(top_days,
  aes(x = factor(Date),
    y = Click_Through_Rate,
    fill = factor(Date))) +
  geom_bar(stat = "identity") +
  labs(
    title = "Top Days by Click-Through Rate",
    x = "Date",
    y = "Click-Through Rate (%)"
  ) +
  theme_minimal()
```



3. Build a table to show the website traffic data for each day. Label the table elements.

```
# Create dataset

traffic <- data.frame(

  Date = c("2023-01-01",
           "2023-01-02",
           "2023-01-03"),

  Page_Views = c(1500,1600,1400),

  Click_Through_Rate = c("2.3%", "2.7%", "2.0%")

)

# Display table

print(traffic)
```

```
Date Click_Through_Rate

1 2023-01-01 2.3

2 2023-01-02 2.7

3 2023-01-03 2.0
```

13. Dataset: Geographic Data

City Population Avg. Temperature Elevation

City A 500000 75 1000

City B 700000 68 800

City C 600000 80 1200

1. Create a map chart to visualize the distribution of cities on a geographic map. Label the map elements.

```
library(ggplot2)
library(maps)

geo$Longitude <- c(-120, -100, -80)
geo$Latitude <- c(35, 40, 45)

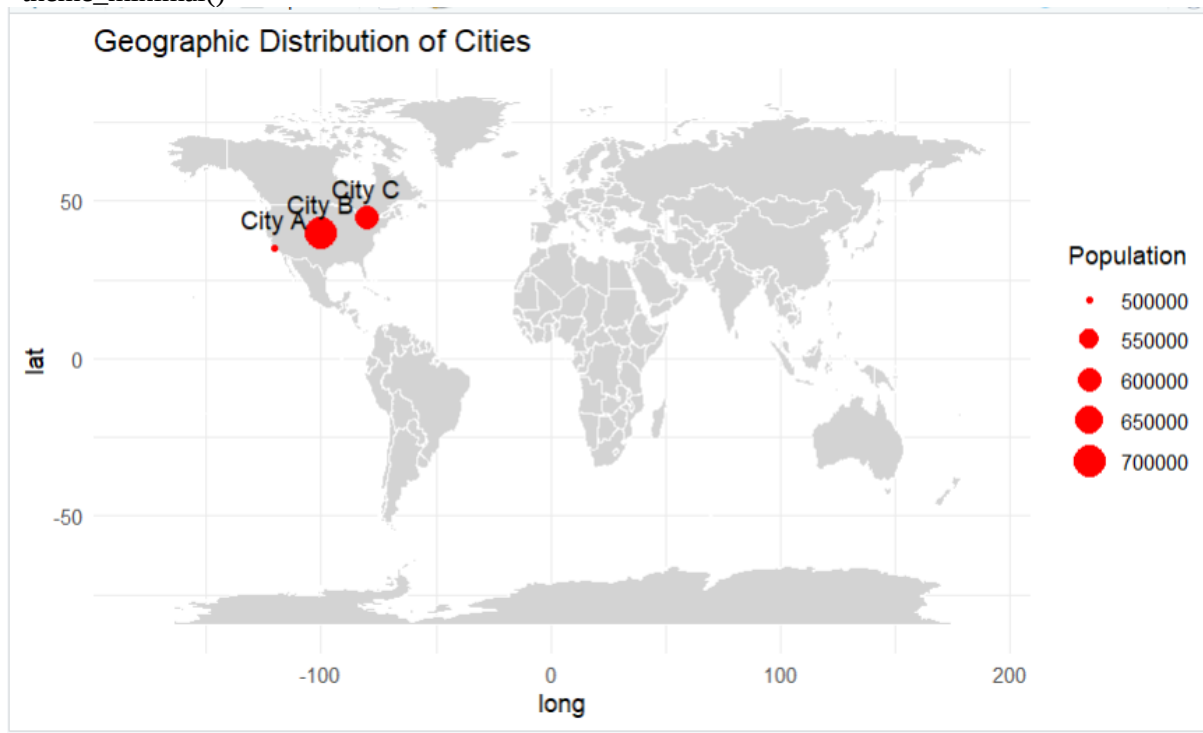
world <- map_data("world")

ggplot() +
  geom_map(
    data = world,
    map = world,
    aes(long, lat, map_id = region),
    fill = "lightgray",
    color = "white"
  ) +
  geom_point(
    data = geo,
    aes(x = Longitude, y = Latitude, size = Population),
    color = "red"
```

```

) +
geom_text(
  data = geo,
  aes(x = Longitude, y = Latitude, label = City),
  vjust = -1
) +
labs(
  title = "Geographic Distribution of Cities"
) +
theme_minimal()

```



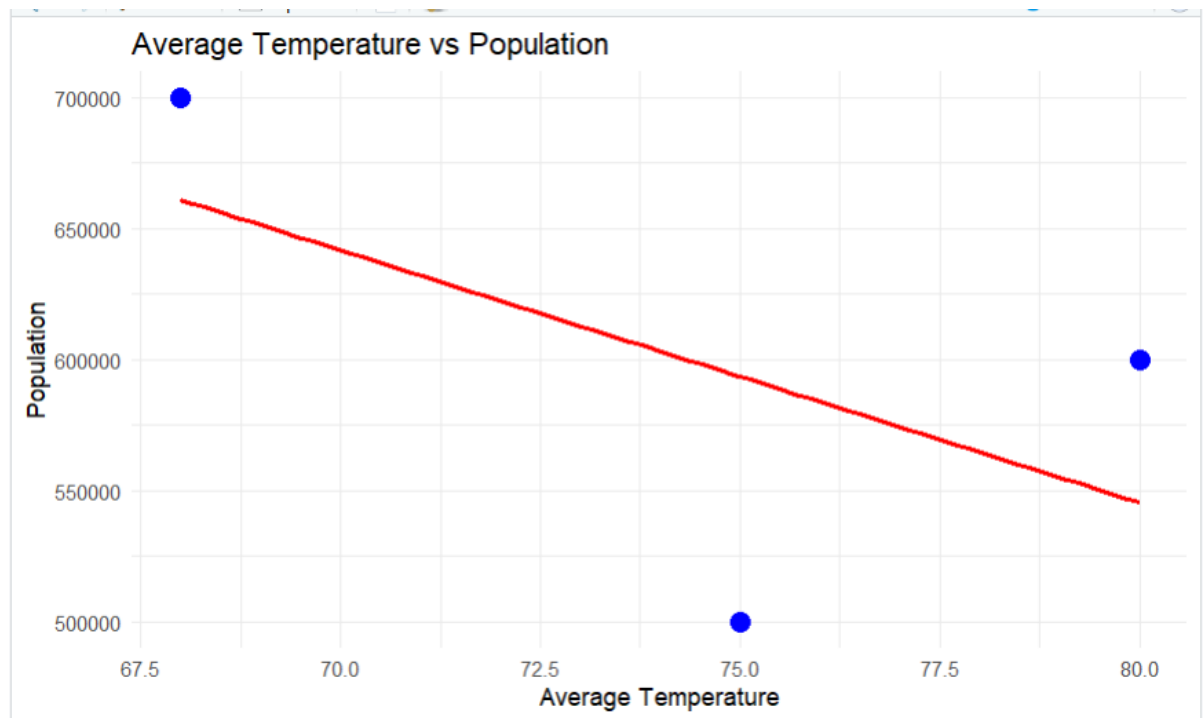
2. Generate a scatter plot to explore the relationship between average temperature and population size. Explain any insights.

```
library(ggplot2)
```

```

ggplot(geographic_data,
  aes(x = Avg_Temperature,
    y = Population)) +
geom_point(
  size = 4,
  color = "blue"
) +
geom_smooth(
  method = "lm",
  se = FALSE,
  color = "red"
) +
labs(
  title = "Average Temperature vs Population",
  x = "Average Temperature",
  y = "Population"
) +
theme_minimal()

```



3. Build a table to display the geographic data for each city. Label the table elements.

Display the geographic data table

View(geographic_data)

	City	Population	Avg_Temperature	Elevation	Longitude	Latitude
1	City A	5e+05	75	1000	-120	35
2	City B	7e+05	68	800	-100	40
3	City C	6e+05	80	1200	-80	45

14. Dataset: Survey Results

Respondent Question 1 Question 2 Question 3

1 A B C

2 B A D

3 C A B

1. Create a stacked bar chart to display the distribution of answers for Question 1. Label the chart elements.

library(ggplot2)

library(reshape2)

Convert data to long format

```
long <- melt(survey,
  id.vars = "Respondent",
  variable.name = "Question",
  value.name = "Answer")
```

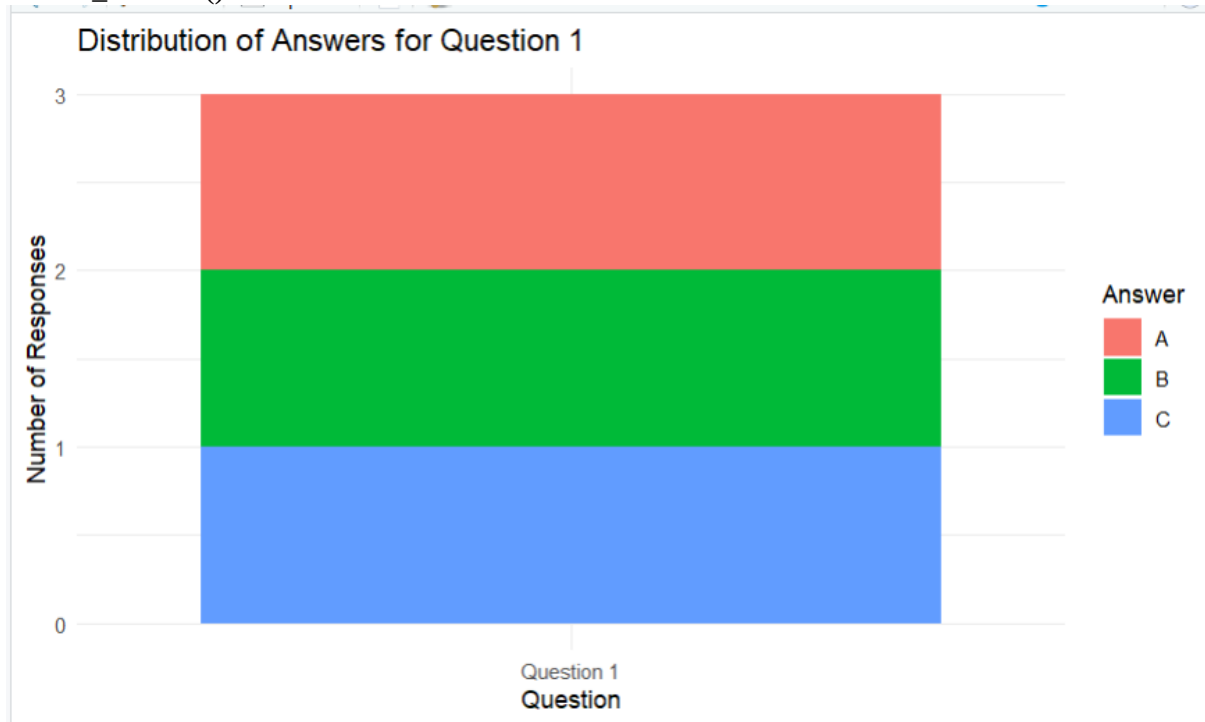
Stacked Bar Chart for Question 1

```
q1 <- subset(long, Question == "Question1")
```

```
ggplot(q1,
  aes(x = "Question 1",
    fill = Answer)) +
  geom_bar() +
  labs(
    title = "Distribution of Answers for Question 1",
    x = "Question",
```



```
y = "Number of Responses"
)+
theme_minimal()
```



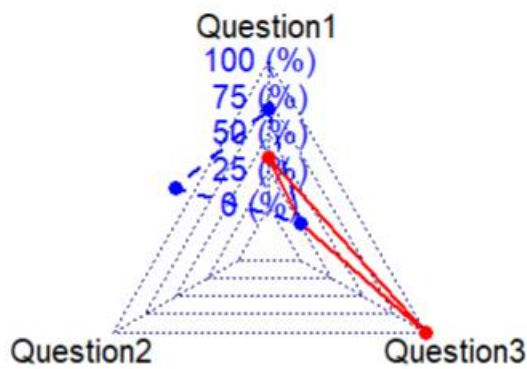
2. Generate a radar chart to visualize the overall pattern of responses across all three questions.

```
score <- data.frame(
  Respondent = survey$Respondent,
  Question1 = c(1,2,3),
  Question2 = c(2,1,1),
  Question3 = c(3,4,2)
)
```

```
radar <- data.frame(
  Question1 = c(4,1,2,3),
  Question2 = c(4,2,1,1),
  Question3 = c(4,3,4,2)
)
```

```
rownames(radar) <- c("Max", "Respondent1", "Respondent2", "Respondent3")
```

```
fmsb::radarchart(
  radar,
  axistype = 1,
  pcol = c("red", "blue", "green"),
  plwd = 2
)
```



3. Build a table to show the survey response data for each respondent. Label the table elements.

View(survey)

or

print(survey)

	Respondent	Question1	Question2	Question3
1	1	A	B	C
2	2	B	A	D
3	3	C	A	B

Set 15: Product Sales Analysis

Dataset: Monthly Product Sales

Product ID Product Name January Sales February Sales March Sales

1 Product A 2000 2200 2400

2 Product B 1500 1800 1600

3 Product C 1200 1400 1100

1. Create a grouped bar chart to visualize the sales of each product for the first quarter. Label the chart elements.

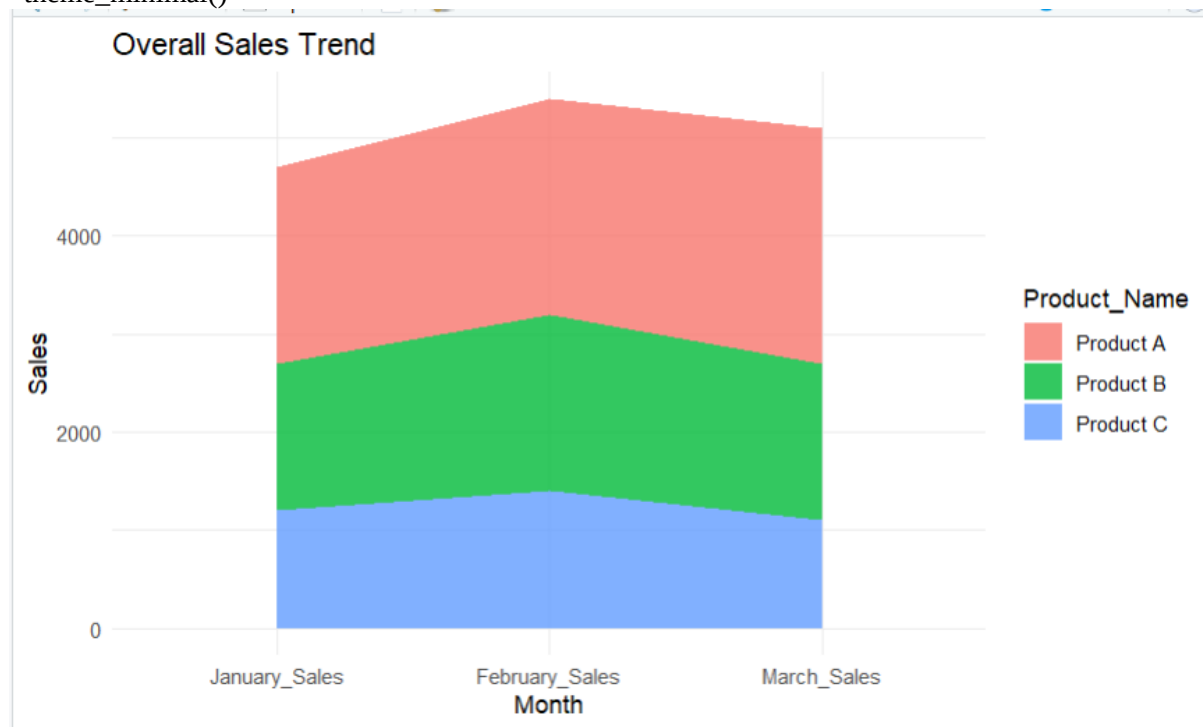
```
long <- melt(
  product_sales,
  id.vars = c("Product_ID", "Product_Name"),
  variable.name = "Month",
  value.name = "Sales"
)
```

```
ggplot(long,
  aes(x = Product_Name,
    y = Sales,
    fill = Month)) +
  geom_bar(stat = "identity",
    position = "dodge") +
  labs(
    title = "Quarter 1 Product Sales",
    x = "Product Name",
    y = "Sales"
  ) +
  theme_minimal()
```



2. Generate a stacked area chart to represent the overall sales trend for all products over the three months.

```
ggplot(long,
  aes(x = Month,
    y = Sales,
    fill = Product_Name,
    group = Product_Name)) +
  geom_area(alpha = 0.8) +
  labs(
    title = "Overall Sales Trend",
    x = "Month",
    y = "Sales"
  ) +
  theme_minimal()
```



3. Build a table to show the monthly sales data for each product. Label the

table elements.

View(product_sales)

OR

print(product_sales)

	Product_ID	Product_Name	January_Sales	February_Sales	March_Sales
1	1	Product A	2000	2200	2400
2	2	Product B	1500	1800	1600
3	3	Product C	1200	1400	1100

5 columns

Product_ID

<dbl>

[1, 3]

0% NA

Product_Name

<chr>

3 unique

0% NA

January_Sales

<dbl>

[1,200, 2,000]

0% NA

February_Sales

<dbl>

[1,400, 2,200]

0% NA

March_Sales

<dbl>

[1,100, 2,400]

0% NA

Showing 1 to 3 of 3 entries, 5 total columns

Set 16: Customer Demographics Analysis

Dataset: Customer Demographics

Customer ID Age Gender Income (in \$)

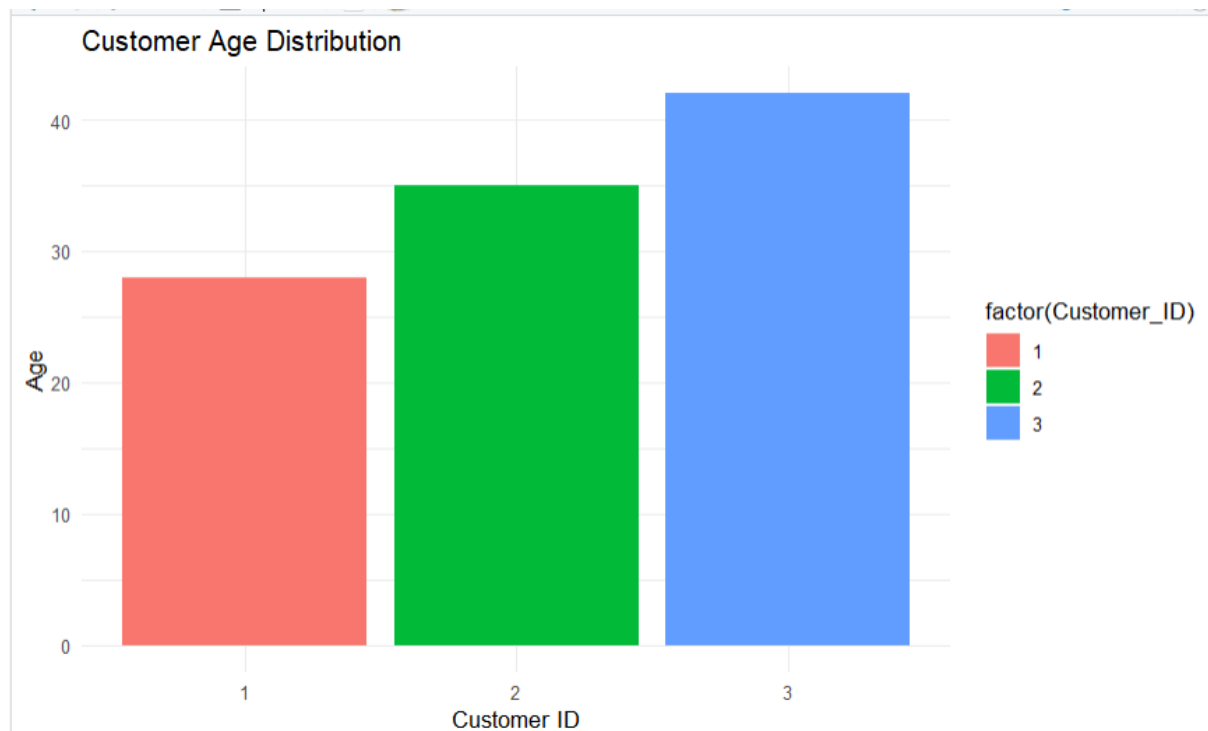
1 28 Female 50000

2 35 Male 60000

3 42 Female 75000

1. Create a bar chart to visualize the distribution of customer ages. Label the axes and title the chart.

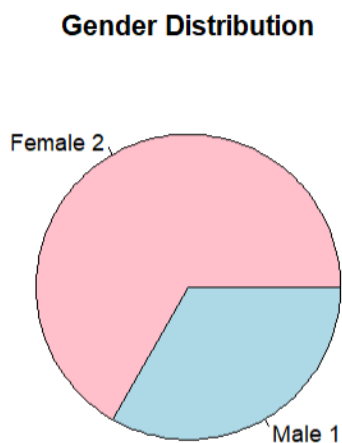
```
ggplot(customer,
  aes(x = factor(Customer_ID),
    y = Age,
    fill = factor(Customer_ID))) +
  geom_bar(stat = "identity") +
  labs(
    title = "Customer Age Distribution",
    x = "Customer ID",
    y = "Age"
  ) +
  theme_minimal()
```



2. Generate a pie chart to represent the distribution of customers by gender.

```
gender_count <- table(customer$Gender)
```

```
pie(
  gender_count,
  labels = paste(names(gender_count), gender_count),
  col = c("pink", "lightblue"),
  main = "Gender Distribution"
)
```



3. Build a table to show the demographic information for each customer.

Label the table elements.

```
View(customer)
```

```
# OR
```

```
print(customer)
```

	Customer_ID ↕	Age ↕	Gender ↕	Income ↕
1	1	28	Female	50000
2	2	35	Male	60000
3	3	42	Female	75000

4 columns ? x	
* Customer_ID <dbl> ▼ ↕	
[1, 3]	0% NA
* Age <dbl> ▼ ↕	
[28, 42]	0% NA
* Gender <chr> ▼ ↕	
2 unique	0% NA
* Income <dbl> ▼ ↕	
[50,000, 75,000]	0% NA

Set 17: Employee Performance Analysis

Dataset: Employee Performance

Employee ID Department Years of Service Performance Score

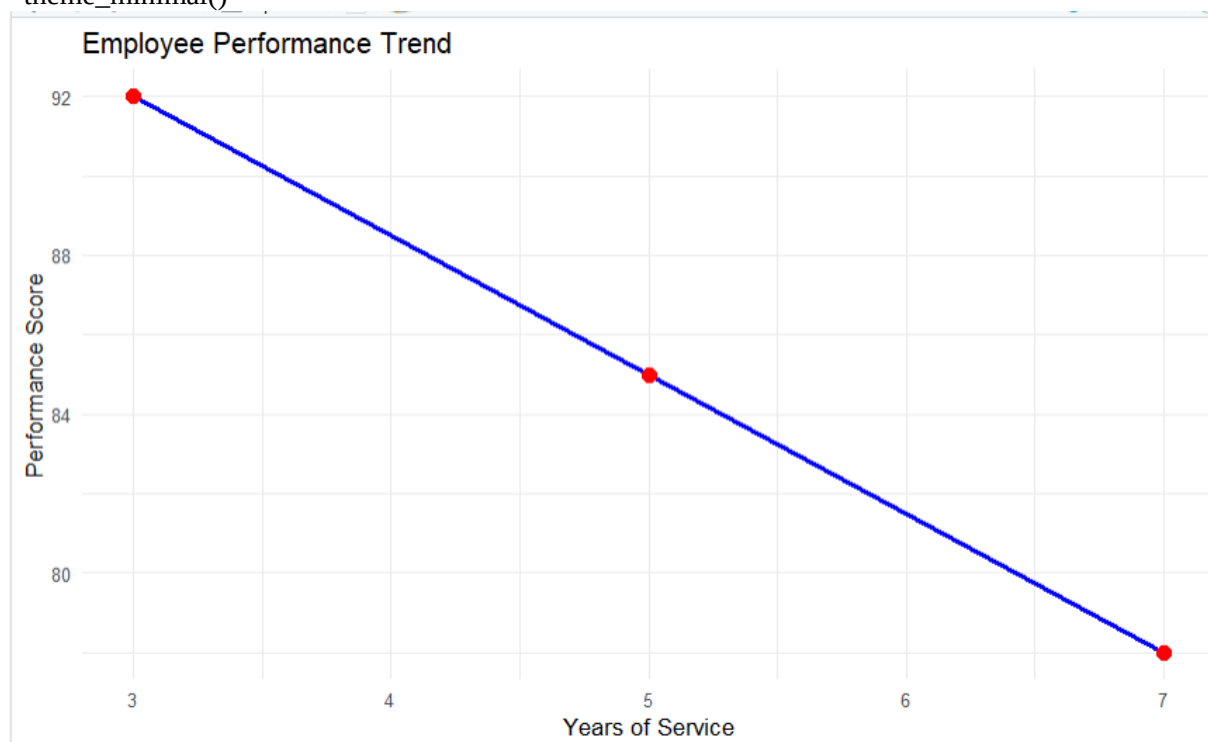
1 Sales 5 85

2 HR 3 92

3 Marketing 7 78

1. Create a line chart to visualize the performance trend of employees over time. Label the axes and title the chart.

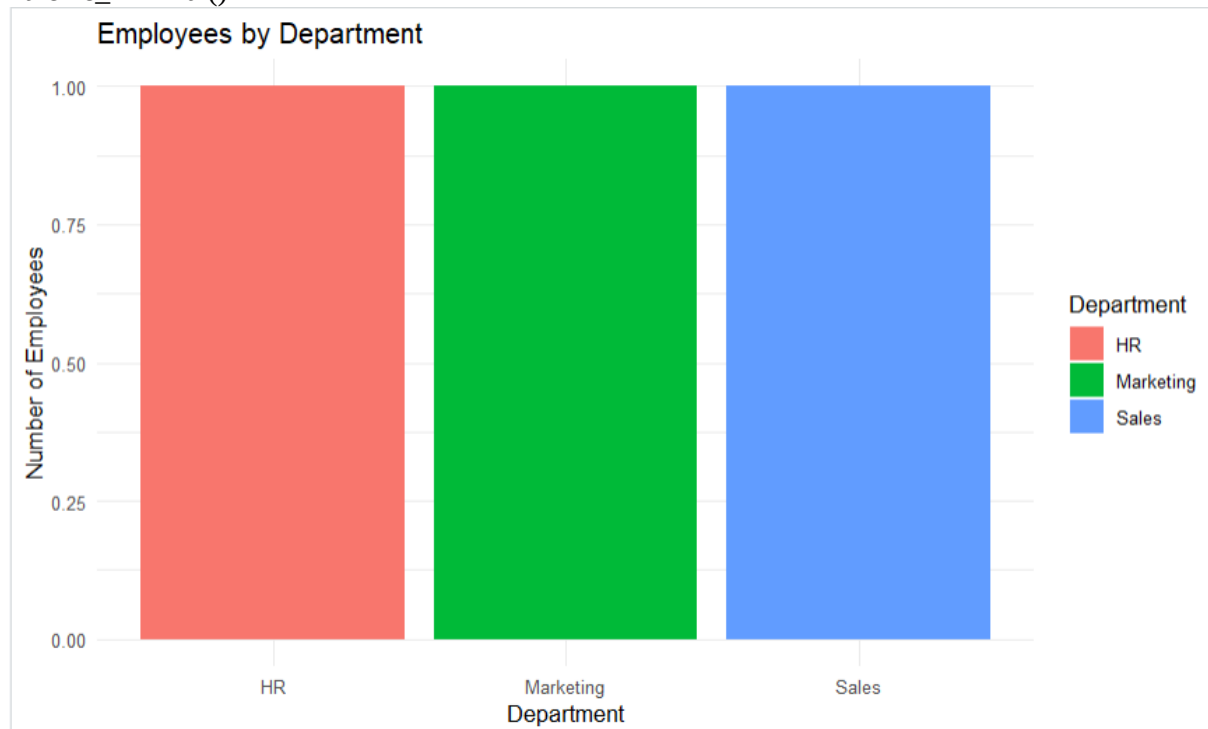
```
ggplot(employee,
  aes(x = Years_of_Service,
      y = Performance_Score,
      group = 1)) +
  geom_line(color = "blue", linewidth = 1) +
  geom_point(color = "red", size = 3) +
  labs(
    title = "Employee Performance Trend",
    x = "Years of Service",
    y = "Performance Score"
  ) +
  theme_minimal()
```



2. Generate a bar chart showing the distribution of employees across

different departments. Label the chart elements.

```
ggplot(employee,
  aes(x = Department,
      fill = Department)) +
geom_bar() +
labs(
  title = "Employees by Department",
  x = "Department",
  y = "Number of Employees"
) +
theme_minimal()
```



3. Build a table to display the performance data for each employee. Label the table elements.

View(employee)

OR

print(employee)

	Employee_ID	Department	Years_of_Service	Performance_Score
1	1	Sales	5	85
2	2	HR	3	92
3	3	Marketing	7	78

4 columns			
Employee_ID <dbl>			
[1, 3] 0% NA			
Department <chr>			
3 unique 0% NA			
Years_of_Service <dbl>			
[3, 7] 0% NA			
Performance_Score <dbl>			
[78, 92] 0% NA			